FPGA Laboratory – Verilog Designs (Experiments 1–5)

Combinational blocks, PicoBlaze, UART, ChipScope and a traffic-light FSM

FPGA laboratory course – B.Sc. Electrical Engineering, Isfahan University of Technology
Author: Kiavash Rabbani Soudabi (student ID 9821663) – term 4012 (Fall 2022) – individual work

Purpose

Five laboratory experiments covering the hardware design flow from combinational logic up to in-system debugging: each one was designed in Verilog, constrained in a UCF file, synthesised and implemented with Xilinx ISE, and accompanied by a written pre-report.

Experiments

- **1 – Combinational building blocks:** half adder, full adder, 2-bit multiplier, 4-bit ripple-carry adder, and a 4-bit multiplier built from shifted partial products.
- **2 – PicoBlaze soft-core:** a hex-to-7-segment decoder driven from the microcontroller output port, exercising the interface between a soft processor and custom logic.
- **3 – UART:** serial communication in two steps – (Q1) a transmitter/receiver pair with a baud-rate clock divider, the transmitter loading its shift register as data plus a start bit, and (Q2) a single UART module with its own transmitter and shift/enable logic.
- **4 – ChipScope in-system debug:** instantiating the Xilinx ICON/ILA cores to capture internal signals on the running device through the JTAG chain, with small stimulus vectors for the capture.
- **5 – Traffic-light finite state machine:** four states implemented with a two-process FSM (next-state logic plus registered state) and an output-decoding process: highway green → highway yellow (after 15 s if the street sensor is active) → street green (after 2 s) → street yellow (after 5 s or when the sensor becomes inactive) → highway green (after 2 s), with a BCD-to-7-segment driver and measured simulation results.

Verification and tools

Verilog-2001 with Xilinx ISE (synthesis, UCF pin/clock constraints, implementation) and ISim for simulation; testbenches are included for the UART and ChipScope experiments, and the FSM simulation results are documented. The submitted pre-reports (experiments 1–4) are part of the repository.

Repository contents

One folder per experiment (`exp1-schematic`, `exp2-picoblaze`, `exp3-uart`, `exp4-chipscope`, `exp5-fsm`) containing the Verilog sources, constraints, testbenches and the ISE project files; `reports/` with the submitted pre-reports; `docs/` with this overview. The experiment folders are reorganised from the original submission, so an ISE project should be pointed at the sources in the folder it is opened from.

Skills demonstrated

Verilog RTL design (combinational, sequential, FSM); hierarchical design and module instantiation; soft-core integration (PicoBlaze); UART protocol implementation including baud-rate generation and framing; in-system debug with ChipScope; display drivers; testbench-based verification; the complete Xilinx ISE flow.